Complete finite search for the rank-three minimum

Human Compression research project

The following is the entire GAP program used in the companion paper. It uses GAP 4.15.1 and SmallGroups 1.5.4. Their group classification and exact group algorithms are retained inputs. The program is needed only for the unrestricted rank-three minimum, not for the uniform construction or the two-group bound.

```
# All groups below the claimed minimum that survive the written order bound.
# SmallGroups classification and GAP exact group algorithms are retained inputs.
L128Check := function()
  local rows,n,id,G,inv,triple,total,pairs,gens;
  rows := [];
  for n in [40,48,56] do
    total:=0;pairs:=0;gens:=0;
    for id in [1..NumberSmallGroups(n)] do
      G:=SmallGroup(n,id);
      inv:=Filtered(Elements(G),g->Order(g)=2);
      for triple in Combinations(inv,3) do
        total:=total+1;
        if not ForAll(Combinations(triple,2),p->Order(p[1]*p[2])=4) then continue;fi;
        pairs:=pairs+1;
        if Size(Subgroup(G,triple))=n then gens:=gens+1;fi;
      od;
    od;
    if gens<>0 then Error("L128_REJECT smaller generating triple");fi;
    Add(rows,[n,NumberSmallGroups(n),total,pairs,gens]);
  od;
  if rows<>[[40,14,3764,0,0],[48,52,19498,88,0],[56,13,9366,0,0]] then
    Error("L128_REJECT exact enumeration totals");
  fi;
  Print("L128_COMPLETE_SMALL_ORDER_SEARCH ",rows,"\n");
end;
L128Check();
QUIT;
```